Fall 2016 / EC571

Instructor:Ajay Joshi

Exam 1

Date: October 4, 2016

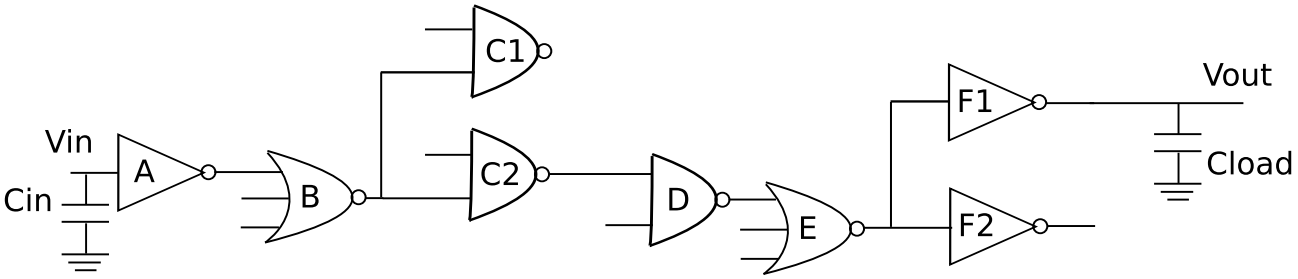
Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. The solutions should be written in the spaces provided on the test sheet.
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
7. Material sharing is not allowed.
8. Calculators are allowed.
9. Boston University Academic Code of Conduct applies.

Question 1



For the circuit shown in the above figure, using the logical effort technique, calculate the input width of transistors for each gate in the path such that total delay of the path is minimized. You can assume that gates C1 and C2 are identical and gates F1 and F2 are identical. Assume Cin = 10 fF, Cg = 1 fF/u, Cload = 120 fF and $2R_{sq(NMOS)} = R_{sq(PMOS)}$. (20 points)

Question 2

Assume the inverters are implemented in standard CMOS with symmetrical VTC. Furthermore, assume $C_{int} = C_g$ ($\gamma = 1$). Equivalent resistance and input capacitance of unit-sized inverter are R_{unit} and C_{unit}, respectively. Sizing factor S ≥ 1. Assume V_{DD} is the same for all inverters.

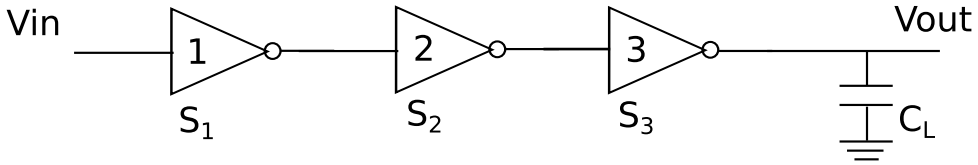


Figure 1: Inverter Chain 1

1. For inverters in the figure above, pick the best sizing factors S_2 and S_3 to minimize propagation delay. What is the minimum delay of the inverter chain (as a multiple of t_{inv})? Assume load capacitance $C_L = 8C_{unit}$ and $S_1 = 1$ which corresponds to an input capacitance of C_{unit} (5 points).

2. What is the total energy drawn from supply when the input switches from 0 to V_{DD}? (Answer in symbolic terms using C_{unit}, V_{DD}) (5 points)

3. For inverters in the figure on the previous page, pick the best sizing S_2 and S_3 to minimize energy consumption. You may assume square wave at the input with period T. What is the total energy consumed for a full cycle i.e. input changes from $0 \rightarrow 1 \rightarrow 0$? (5 points)

4. What is the delay (in terms of t_{inv}) of the circuit in the figure below (5 points)?

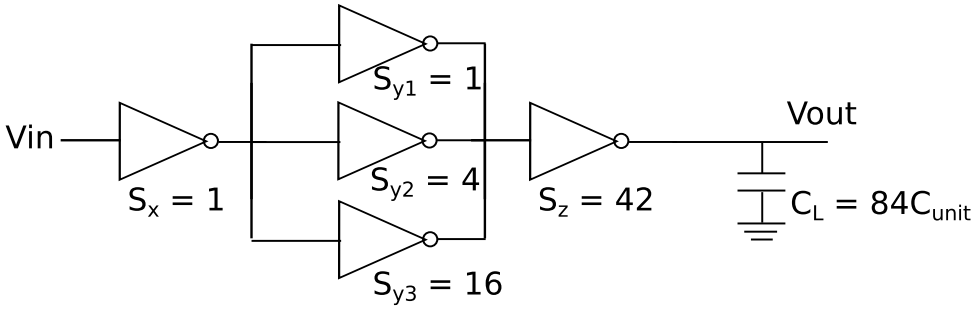


Figure 2: Inverter Chain 2

Question 3

Your friend claims that the circuit shown in the figure below is a digital logic gate. Determine the values of V_{out} for all 4 possible input combinations (A = 0, B = 0; A = 0, B = 1; A = 1, B = 0; A = 1, B = 1). What logic function is implemented by the the circuit shown below? Use L = 100 nm, $V_{TH,n}$ = 0.4 V, $V_{TH,p}$ = -0.4 V, $R_{sq(NMOS)}$ = 10 $kohm/sq$, $R_{sq(PMOS)}$ = 20 $kohm/sq$, μ_n = 400 $cm^2/(V.s)$, μ_n = 200 $cm^2/(V.s)$, C_{ox} = 1.125 $\mu F/cm^2$ and v_{sat} = 1e7 cm/s.

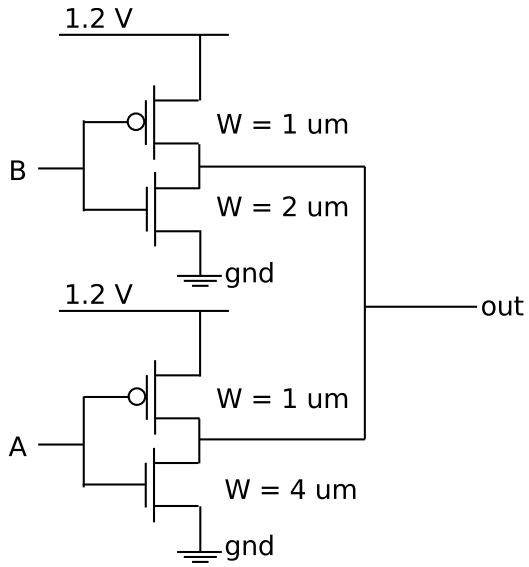


Figure 3: Digital gate

Question 4

1. Draw the static complementary CMOS logic implementation for the following function - $f(A,B,C,D) = \overline{A + B \cdot C + D}$. (6 points)

2. Determine the widths of the various transistors as a function of W so that worst case $\tau_{HL} = \tau_{LH}$ and the delay matches with the delay of an inverter. You can assume $2.5R_{sq(NMOS)} = R_{sq(PMOS)}$. Ignore the internal capacitances of the non-output nodes while determining the transistor sizing.(7 points)

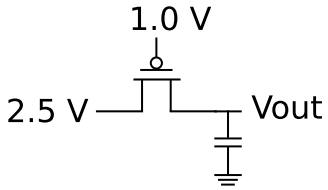
3. Calculate the worst-case τ_{HL} and τ_{LH} for both gates using the following parameters – width of the NMOS in the inverter W = 1.0 micron, L = 0.25 micron $R_{sq(NMOS)}$ = 10 Kohm/sq, $R_{sq(PMOS)}$ = 25 Kohm/sq, C_g = 1.2 fF/micron, C_d = 1.5 fF/micron, C_L = 100 fF. (7 points)

Question 5

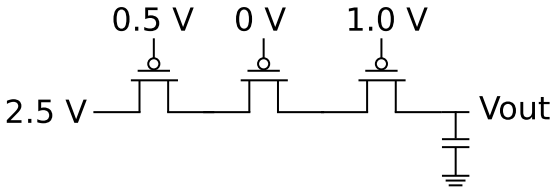
Design the static complementary CMOS implementation of a 2-bit comparator circuit, where we have two inputs A and B (each is 2-bit wide) and the output = 0 if A $>$ B and output = 1 if A $\leq$ B. Design the circuit for minimum delay (assuming a stage effort of 4) and driving a load of 10 fF. As part of the design you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron, minimum transistor width = 0.3 micron, $R_{sq(NMOS)}$ = 13 Kohm/sq, $R_{sq(PMOS)}$ = 26 Kohm/sq, C_g = 1 fF/micron, C_d = 1.5 fF/micron. You can assume that you have both complemented and non-complemented inputs and that input signals have very small rise and fall times. *Hint: Determine the Boolean expression of the comparator circuit and use the Boolean expression to design the circuit. Use logical effort technique to determine the transistor sizes.* (20 points)

Question 6

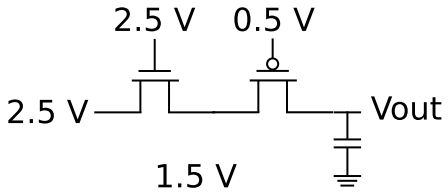
Find the final value of the voltage V_{out} for the following 5 configurations (4 points each). Assume $V_{th,N}$ = 0.5 V and $V_{th,P}$ = -0.5 V. Assume that the capacitor is initially discharged. Ignore subthreshold conduction and body effect.



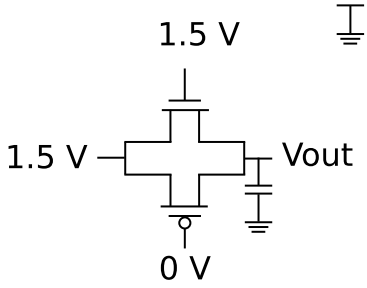
Vout =



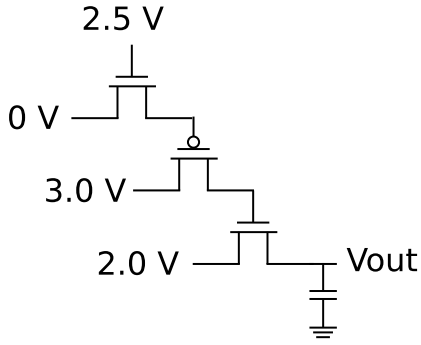
Vout =



Vout =



Vout =



Vout =

Bonus question (5 points)
Prof. Joshi wanted to go to Rio Olympics this year, but could not due to other commitments. Please list 5 Olympic sports that he should have definitely attended if he had gone to Rio Olympics.